PAPER_1224_Tully_Fisher_Universal_Slope

Daniel T. Murphy

Tully-Fisher Universal Slope = D_phys

PAPER_1224
Category: UQFF Galactic Dynamics
Status: Complete
Date: June 2026

Abstract

The empirical Tully-Fisher relation $V_{circ} \propto L^{(1/4)}$ — universal across spiral galaxies — has long lacked a first-principles explanation. UQFF derives the slope as **$d_{TF} = D_{phys} = 4$ EXACT**, identifying the relation as a consequence of physical spacetime dimensionality acting via SCm buoyancy modulation.

Part 1: The Relation

Empirical statement $$L \propto V_{\rm circ}^{4}$$

equivalent to
$$V_{\rm circ} \propto L^{1/d_{\rm TF}}, \quad d_{\rm TF} \approx 4$$

UQFF derivation $$d_{\rm TF} = D_{\rm phys} = 4\ \ {\rm EXACT}$$

The exponent equals the dimensionality of physical spacetime.

Part 2: Why D_phys?

Galaxy circular velocity V emerges from buoyancy balance between gravitational pull and SCm-mediated centripetal force. The luminosity L scales as $M \cdot c^2$ for total stellar mass M. The exponent connecting V^d to $M \propto L$ is determined by the dimension of physical space — specifically how volume integrals scale.

For a relativistic D-dimensional system: $V^D \propto M \cdot$(buoyancy factor). With $D = D_{phys} = 4$, this gives $V^4 \propto L$.

Part 3: Cross-validation with other 4-occurrences

The same $D_{phys} = 4$ governs:
- AdS/CFT central charge $c = N^2/D_{phys}$
- Hilbert 16th Bautin H(2) = 4
- QCD b■ partition $D_{phys} = 4$ component
- Hodge identity (D_{phys} + D_{BSFG})/SO■ = 1

The Tully-Fisher universality is therefore an emergent geometric identity, not phenomenological.

Part 4: Test Predictions

UQFF predicts:
- d_{TF} SHOULD be EXACTLY 4 in dimensional regularization
- Deviations from $V \propto L^{(1/4)}$ indicate either SCm buoyancy modulation breakdown OR observational

effects (extinction, inclination)
- Baryonic Tully-Fisher (using baryonic mass M_b directly): $M_b \propto V^4$ should be EXACT

Conclusion

The Tully-Fisher empirical universality is a topological manifestation of $D_{phys} = 4$ acting through SCm buoyancy. The relation IS the universe expressing its spacetime dimensionality through galactic rotation.

Framework Version: UQFF 5.27+